

Ivana Bajakić
Marta Božina Beroš
Editors

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**EU FINANCIAL
REGULATION
AND MARKETS**

**Beyond Fragmentation
and Differentiation**



PRAVNI FAKULTET SVEUČILIŠTA U ZAGREBU



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Ivana Bajakić
Marta Božina Beroš
Editors

EU FINANCIAL REGULATION AND MARKETS
Beyond Fragmentation and Differentiation

Zagreb, 26-27 November 2020

Conference Proceedings

Faculty of Law University of Zagreb

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Beyond Fragmentation and Differentiation

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AN OVERVIEW OF REGULATORY STRATEGIES ON CRYPTO- ASSET REGULATION - CHALLENGES FOR FINANCIAL REGULATORS IN THE WESTERN BALKANS

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ABSTRACT

The rise of the crypto-asset market has opened up a number of questions on their benefits and risks. As a new form of virtual property, cryptocurrencies and tokens of investment nature are characterized by specific technological infrastructure. In order to understand the regulatory perspective of the crypto-asset market, it is necessary to briefly present the issues related to digital assets infrastructure and the forms of crypto-assets, which blur the line between financial products and virtual property. Crypto-assets represent a form of fintech innovation that could materially affect the financial landscape (notably payments, investments and capital raising), may have impact on the financial sector and therefore may create conditions for regulatory arbitrage where regulated participants operate in a highly regulated environment. The Paper aims to identify the basic issues of the crypto-assets regulatory framework and the extent of applicable regulatory approaches. Comparative regulatory practices of selected countries, as well as the global regulatory perspective of this market, serve as a guide to assess the existing regulatory framework and regulatory challenges, in order to assess urging issues which the regulators in Western Balkan countries are facing. Regulators are confronted with a dilemma: how to promote financial innovation while preserving financial stability and protecting investors. That is why many regulators have adopted a regulatory stance in regulating distributed ledger (blockchain) technologies and tend to formulate regulatory strategies which are risk based, phased and adaptive. Regulatory responses vary from a complete ban on issuing or trading crypto-assets, warnings and principles-based regulation, to extending existing rules on the capital markets and payment systems to specific intermediaries which are licensed almost as financial institutions per se. A tendency towards developing a new set of legislation specifically aiming to register and/or license providers of services connected to virtual property based on cryptography may be observed, and a proposal for an EU regulation on Markets in Crypto-Assets is a clear example.

Keywords: crypto-assets, virtual currencies, blockchain technology, fintech, financial regulation

1. INTRODUCTION

The increase in the number of virtual currencies and other forms of virtual assets is accompanied by an increase in the volume of trading and the use of the new

technology on which they are based. Regulators' concerns about the risks arising from the use of new technologies and the wider use of crypto-assets are growing. Regulators around the world face the challenge of achieving balance between the need to regulate a new phenomenon and stifle innovation, or fostering innovation and taking a balanced regulatory stance with regards to the new technologies.

Whereas some already existing risks in the financial system are an inherent characteristic of modern forms of financial intermediation (e.g. risk of use for money laundering purposes), the use of new technologies raises questions of new risks that are specific to digital property. Crypto-assets represent one item among various technological advancements which drive the change in the modern financial sector. The underlying blockchain/ digital ledger technology (DLT) by itself creates a number of problems that regulators face in defining the appropriate regulatory strategy. Therefore, setting the regulatory agenda should be based on the analysis on the risks and the most important legal challenges arising from that technology, as well as the basic infrastructure elements of the crypto-asset industry. One of the key issues is the following: to what extent the existing regulations applicable to financial instruments may be applicable to emerging forms of new assets?

States, national and international regulators are striving to encourage the development of innovative technologies, while on the other hand facing public expectations to protect consumers and investors, businesses participating in this market, and the financial system. Many countries react gradually or adopt the "wait and see" approach, to get some extra time to consider how other countries react to crypto-evolution.¹ Because the nature of international finance is global, national and regional initiatives have limited maneuver for intervention. The basic problem national regimes are facing is the dispersed or distributed architecture of DLT based systems underlying crypto-assets. Reaching a consensus at the international level is a tantamount to developing national regulatory responses. The fragmented nature of FinTech tools dictates the emergence of at least the minimum requirements of a transnational regime with regard to the most important risks. An excellent example of transnational

¹ Wulf A. Kaal, 'Initial Coin Offerings: The Top 25 Jurisdictions and Their Comparative Regulatory Responses' (2018) 1 *Stan J Blockchain L. & Pol'y* 41, 63.

initiatives to combat money laundering and terrorist financing are the recommendations of the Financial Action Task Force on virtual currencies.²

The regulatory response or the approach to the regulation of the crypto asset market from a public law perspective, occupies a primary place in this paper. It explores the basic issues of a normative nature that are introduced to help in managing risks stemming from the use of new financial technologies in issuing and providing services related to trading in crypto-assets. Public law regime sets out minimum standards for establishment, operation, internal organization and risk management, as well as requirements on fair dealing, customer care and consumer protection. In that sense, issues related to private law aspects, e.g. the issuance, management and servicing of claims on virtual property, mutual relations of transactors, as well as issues of criminal liability for abuse in the virtual currency market have not been explored, and neither have tax policy issues.

The Paper explores the basic settings of public regulation of the market in crypto-assets and the main principles for an appropriate choice of a regulatory strategy. It does not aim to portray a comparison of the existing regulations. The basic principles of crypto-asset regulation and regulatory approaches to regulation will be portrayed, with a special reference to alternatives to traditional "command and control" regulation. Innovation in the financial sector implies not only adjustments and innovations in rules, but also approaches to regulation. This is particularly challenging for the countries of the Western Balkans, where the focus has long been on firm rules and a rigid concept of the role of regulators. The final part of the paper will focus on addressing the challenges that legislators and regulators in the Western Balkan region are facing in developing an approach to crypto market regulation.

2. CRYPTO-ASSET IN THE REALM OF FINTECH AND RECONCEPTUALIZATION OF REGULATION

The past decade was marked by and unprecedented rise of a new type of financial intermediaries and providers, often referred to as "*fintech*", which offer innovative services. In the broader sense, fintech refers to "incremental or disruptive innovations in or in the context of the financial services industry

² FATF, Guidance for a risk-based approach 'Virtual assets and virtual asset service providers', June 2019,

<<https://www.fatf-gafi.org/publications/fatfrecommendations/documents/guidance-rba-virtual-assets.html>> accessed 15 August 2020

induced by IT developments resulting in new intra- or inter-organizational business models, products and services, organizations, processes and systems".³ Unlike traditional financial institutions, often finance giants, innovative firms as a rule are smaller players, predominantly focused on competitive and innovative products.

The digital setting and use of artificial intelligence challenged the incumbent financial infrastructure. Financial intermediation has shifted from conventional banks to non-depository financial institutions that do not have to comply with detailed prudential and conduct of business rules. Fintech is therefore able to avoid intermediation costs and minimum capital requirements. Fintech stimulates financial sector development, and is pivotal in increasing the diversity and accessibility of financial services.⁴ Crypto-assets are a form of fintech innovation that could materially affect the financial landscape (notably payments, investments and capital raising), may have impact on the financial sector and may create conditions for regulatory arbitrage where regulated participants operate in a highly regulated environment.

Regulation usually goes behind technological innovation and is often tardy in responding. Elevated complexity of the financial landscape brought new challenges for the regulatory framework and urged regulators to react, as the current legal framework is unable to respond to such challenges. The intersection of technology and functional powers of the fintech industry, emerging business models, novel services and consumer expectations, challenged the regulators and in many different ways reconceptualized financial regulation.⁵ One of the main issues in the context of fintech regulation is regulatory uncertainty, and therefore the fundamental question is how to create regulatory certainty and provide flexibility to react to unknown risks and novel business models. Overly precautionary approach, prohibition of certain types of fintech and premature changes in hard law may impede innovation and make national financial system uncompetitive. The challenge is thus to design a regulatory environment that is flexible enough to accommodate new fundamental changes to markets and, at the same time, is able to create

³ Thomas Puschmann, 'Fintech' (2017) 1 Business & Information Systems Engineering 59, 69, at 74.

⁴ Daniela Gabor, Sally Brooks, 'The digital revolution in financial inclusion: International development in the fintech era' (2017) 4 New Pol Ec 22, 423.

⁵ Douglas W. Arner, Janos Barberis, Ross P. Buckley, 'FinTech, RegTech, and the reconceptualization of financial regulation' (2017) 3 NW J Int'l L & Bus 37 371.

regulatory certainty for all market participants.⁶ This has led to the reconceptualization of regulation and innovations in the regulatory process reflected in a more flexible approach based on the application of principles, rather than hard law, as well as innovations in the process of regulation in the form of testing environments based on dialogue, enabling regulators to decrease uncertainty through learning.

Most laws on financial intermediation were written before rising fintech innovations occurred.⁷ Rules may conflict with the new setting of innovative means and some fintech may not fit categories formerly determined by applicable laws. Moreover, some innovations prompted structural changes and regulators must firstly assess those risks and determine whether they are encompassed by the existing framework. The sooner regulators are introduced to innovation, the better they will identify risks, whereas financial innovators will get the opportunity to adapt their service while costs of compliance are still reasonable.

3. THE CORE RATIONALES FOR REGULATING CRYPTO-MARKETS

3.1. The risk of new technologies and regulatory adaptation

Cryptocurrencies, as the first forms of crypto-assets, appeared in 2008, when Bitcoin was created as a specific electronic payment system without traditional financial intermediaries to guarantee that the transaction will be completed.⁸ Through a local network of Internet-connected computers ("peer-to-peer network"), in which participants can see all transaction data at any time (without participant identification), participants agree to verify the chronology of transactions by solving complicated cryptographic algorithms, which form blocks of time-grouped transactions. Such a chain of block is visible in a publicly available database of unique transaction history. In this sense, blockchain technology represents the documentation of records of all transactions, grouped into blocks and provided by cryptography, within a decentralized alternative electronic system, which takes the form of a publicly available fragmented general ledger. Hence it is also called distributed ledger

⁶ Wolf-Georg Ringe, Cristopher Ruof, 'Regulating Fintech in the EU: the Case for a Guided Sandbox' (2020) 3 EJRR 11 604, 613.

⁷ Mark Fenwick, Erik P Vermeulen, Wulf Kaal, 'Regulation Tomorrow What Happens When Technology is Faster than the Law' (2017) 3 Amer Uni Bus LR 6 561.

⁸ Satoshi Nakamoto, *Bitcoin: A peer-to-peer electronic cash system* (2008) <<https://bitcoin.org/bitcoin.pdf>> accessed 20 September 2020

technology (DLT).⁹ Somewhat later, new forms of transaction verification have been developed. Instead of mining (the process of solving algorithms that create a chain of blocks), new technological innovations enabled verification of transactions among parties ("proof-of-stake"), on the basis of automated decisions and smart contracts. The development and application of technological innovations through Decentralized Autonomous Organizations (DAO) altered the world of alternative finance.¹⁰

Although initially used in cryptocurrency schemes, blockchain technology has a much wider potential for application because it allows various documents to be encrypted, including those whose existence can be proven in secure publicly distributed books. Many potential uses of this technology raises an important question: whether the transfer of value without the intervention of banks and central authorities that the state can control implies a specific new governance model?¹¹

The theory proposes to recognize a new type of right over virtual property.¹² De Filippi and Wright believe that these autonomous systems of rules based on codes create an order without law, something that could be called private regulatory frameworks, which they called the *lex cryptographica*. Such frameworks allow people to communicate, organize and exchange value without the participation of intermediaries, centralized bodies and the state.¹³ Public

⁹ Government Office for Science, *Distributed ledger technology: beyond block chain*, A report by the UK Government Chief Scientific Adviser (2015), 36. <https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/492972/gs-16-1-distributed-ledger-technology.pdf> accessed 20 September 2020.

¹⁰ Garrick Hileman, Michel Rauchs, *Global Cryptocurrencies Benchmarking Study*, University of Cambridge, Centre for Alternative Finance (2017) <<https://cointelegraph.com/storage/uploads/view/2017-global-cryptocurrency-benchmarking-study.pdf>> accessed 20 September 2020.

¹¹ Primavera De Filippi, Benjamin Loveluck, 'The invisible politics of Bitcoin: governance crisis of a decentralised infrastructure', (2016) 3 *Internet Pol Rev* 5 1 <<https://policyreview.info/articles/analysis/invisible-politics-bitcoin-governance-crisis-decentralised-infrastructure>> accessed 15 August 2020; Aaron Wright, Primavera De Filippi, 'Decentralized Blockchain Technology and The Rise of Lex Cryptographica' (2015) <<http://ssrn.com/abstract=2580664>> accessed 25 September 2020.

¹² Carla L. Reyes, 'Moving Beyond Bitcoin to an Endogenous Theory of Decentralized Ledger Technology Regulation: An Initial Proposal' (2016) 61 *Vill L Rev* 191, 193; Katie Szilagyi, 'A Bundle of Blockchains? Digitally Disrupting Property Law' (2018) 48 *Cum L Rew* 9, 24.

¹³ Wright and De Filippi (n 11).

blockchain networks enable creation and functioning of rule based systems which are forerunners of new institutional forms of economic governance based on rule-system for economic coordination between various actors.¹⁴ Blockchain systems can be controlled in areas where they intersect with regulated entities such as network operators that communicate with a blockchain-based system and all those who develop or support technology. This is one of the basic hypotheses of crypto-asset regulation.

Given that blockchain technology is still in development, there is a danger that premature and excessive regulation may prevent the emergence of novel applications of technology. Premature introduction of a licensing system and control could prevent participants in this market from freely experimenting with new technology, which essentially hampers innovation. At the same time, the lack of regulations would leave those who want to apply the technology in a gray area, where they are not sure if what they are doing is legal. Applied to the crypto-asset market, this paper will cast a light on innovations in the regulatory approach. Considering that virtual currency schemes are based on technology the use of which may circumvent financial intermediation, its postulates such as aggregation and netting, the role of financial institutions as traditional intermediaries seems to be somewhat diminished. It certainly poses a potential risk to financial stability. The functioning of the classic payment system could be jeopardized if a larger number of users of virtual currency schemes appeared. However, the risk posed by innovation can encourage financial institutions to adapt to new technologies. Central banks and financial regulators are increasingly noticing the need to adjust regulations and enable the so-called tokenization of securities through the procedure of initial public offering of coins.¹⁵ The legal framework should also ensure technological neutrality with

¹⁴ Sinclair Davidson, Primavera De Filippi, Jason Potts, *Disrupting governance: The new institutional economics of distributed ledger technology*, SSRN Working Paper Series (2016) at 6-7, < https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2811995 > accessed 20 October 2020.

¹⁵ For instance, France had taken the lead as early as 2016, with the Ordonnance n°2016-520 of April 28th 2016

("Minibons" *ordonnance*) setting out the conditions for using "shared electronic recording devices". Ordonnance was amended by the Decree of December 24th 2018 (known as the "Blockchain Decree"). In Luxembourg, the Law of 1 March 2019 amending the Law of 1 August 2001 concerning the circulation of securities and other financial instruments, provides that securities can be held using distributed ledger technologies and these technologies can also be used to register transfers (Loi du 1er mars 2019 portant modification de la loi modifiée du 1er août 2001 concernant la circulation de titres).

regard to various protocols on which the infrastructure of the blockchain may be based. The flexible approach through technological neutrality principle is necessary to differentiate a legal regime from rapidly evolving technologies.

3.2. Risks related to the stability of financial system and public order

Numerous reasons on the supply and demand side of virtual currencies have contributed to their development.¹⁶ Digital currencies based on the use of DLT represent the evolution of payment systems, with reduced transaction costs and increased speed, including in the areas of e-commerce and cross-border transactions. The success of Bitcoin and other cryptocurrencies has motivated the emergence of start-up financing through Initial Coin Offerings as an alternative to debt and capital market instruments. However, crypto-assets also enable the financing of companies which "bypass" the regulations that regulate financing on the capital market, which provide certain legal protection to investors and issuers. The immediate character of transactions, i.e. the absence of intermediaries and the direct execution of transactions with the use of keys, allows greater discretion compared to other payment systems. Due to insufficient regulation, these transactions were of interest to owners of suspicious capital and were used to finance criminal activities, which is one of the main reasons for regulating virtual currency schemes.¹⁷ Examples of cryptocurrency fraud and abuse are numerous: hacker attacks, money laundering, terrorist financing, crypto extortion, theft, abuse of women and children, Ponzi schemes, phishing, scams, fake sites, etc.¹⁸

The lack of a regulatory framework allows for "circumvention" of regulations and makes it difficult to implement official public policies. The most obvious example is tax avoidance, as tracking and identifying transaction participants is difficult, these virtual currencies represent new "super tax havens." Also, the free exercise of cross-border payments may be in conflict with foreign exchange regulations or regulations restricting capital flows. The lack of regulation places

¹⁶ Bank for International Settlements, Committee on Payments and Market Infrastructures, *Digital Currencies*, (2015), 7-12

<<https://www.bis.org/cpmi/publ/d137.pdf>> accessed 15 August 2020.

¹⁷ Lawrence C, 'Virtual Currencies; Bitcoin & What Now After Liberty Reserve, Silk Road, and Mt. Gox?' (2014) 20 Rich. J. L. & Tech 4, 1, 3; Tara Mandjee, 'Bitcoin, Its Legal Classification and Its Regulatory Framework' (2014) 15 J. Bus & Sec. L 2 157.

¹⁸ Homeland Security Studies and Analysis, *Risks and Threats of Cryptocurrencies*, Publication Number: RP14-01.03.03-02 (2014),

<[https://www.anser.org/docs/reports/RP14-01.03.03-](https://www.anser.org/docs/reports/RP14-01.03.03-02_Cryptocurrencies%20508_31Dec2014.pdf)

02_Cryptocurrencies%20508_31Dec2014.pdf> accessed 15 August 2020.

cryptocurrency owners mostly outside the reach of national fiscal and monetary policy, which can make their implementation more difficult and negatively affect the efficiency of economic policy.

Central banks closely monitor the development of the cryptocurrency market, as well as the development and application of new technologies that have the potential to reshape the world of finance, especially in terms of achieving their basic goals and objectives.¹⁹ The growth of the cryptocurrency market could lead to the displacement of conventional currencies. This would raise the question of the monetary risks of using cryptocurrencies, since their supply does not depend on the monetary authorities, unlike the supply of traditional currencies. Impossibility of the central bank to influence the flow of assets could cause a loss of confidence in the national currency and prevent the central bank from carrying out its basic monetary function and effectively performing the role of lender of last resort. The potential of cryptocurrencies to affect financial stability depends on their connection to the real economy and financial markets. Given that they have not yet reached a critical volume of trade and a wider circle of users, the prevailing view is that they cannot affect financial stability.²⁰

The emergence of new services related to the application of new technologies opens new risks beyond the control of central banks and regulatory bodies, which can lead to the loss of confidence in the financial system. Recorded cases of collapse of cryptocurrency exchanges and theft from electronic wallets clearly indicate dangers for investors and consumers. Encompassing different forms of assets and intermediaries, crypto-markets are prone to various different legal risks.²¹ Emerging forms of cryptocurrencies carry various risks from the point of view of consumers, starting from the fact that they are unsuitable products, i.e. that due to their characteristics they are not adapted to consumers. Consumers

¹⁹ International Monetary Fund, *Virtual Currencies and Beyond: Initial Considerations*, IMF Staff discussion note, SDN/16/03 (2016), 31-33; Gina C. Pieters, "The Potential Impact of Decentralized Virtual Currency on Monetary Policy," Fed Res Bank of Dallas, Globalization and Monetary Policy Institute, 2016 Annual Report (2017);

²⁰ Financial Stability Board, *Crypto-asset markets, Potential channels for future financial stability implications*, (2018); <<http://www.fsb.org/wp-content/uploads/P101018.pdf>> accessed 10 October 2020; Financial Stability Board, FSB Chair's letter to G20 Finance Ministers and Central Bank Governors (2018), <<http://www.fsb.org/2018/03/fsb-chairs-letter-to-g20-finance-ministers-and-central-bank-governors/>> accessed 10 October 2020.

²¹ European Central Bank, *Impact of digital innovation on the processing of electronic payments and contracting: an overview of legal risks* (Frankfurt am Main 2017) <<https://www.ecb.europa.eu/pub/pdf/scplps/ecb.lwp16.en.pdf?3f7054a2a21c98ba0560ca4dc8ff329d6>> accessed 15 August 2020.

are often unaware of their risks or do not have enough information to make a rational decision, and fraudulent practices as well as various forms of market manipulation are common.²² It is a difficult task for an average consumer to determine the value of investing in crypto-assets. Given that their price is primarily based on speculation that consumers often overestimate the expected profit without paying attention to what is necessary to consider when assessing investment risk.²³ A special type of consumer abuse or price manipulation are the so-called pump and dumps, where the price and value of a crypto-asset is systematically increased by a collective buying of that crypto-asset, followed by the abrupt sale of that crypto-asset at the increased price. The overemphasis on benefits and lack of information on potential losses especially characterizes the IPOs of coins and other tokens, where unrealistic expectations of success and risks are often presented to consumers in the form of a shortened prospectus (White Paper), which makes it impossible to understand the risk profile. Some authors claim that cryptocurrencies often represent a form of fraudulent practices, comparing them to financial bubbles, and therefore suggest limited access to these schemes for a wide range of investors, primarily consumers as non-professional investors.²⁴

4. SETTING THE SCENE: BASIC ISSUES IN DEFINING THE REGULATORY STRATEGY

4.1. The key principles and forms of crypto-assets regulation

Up to date, national regulators responded in different ways ranging from the "wait-and-see" approach, issuing warnings, to sanctioning the avoidance of conforming to capital market rules. Regulatory response was mostly concerned with illicit transactions, market integrity and customer protection.²⁵ Despite the activities undertaken in individual countries, where regulatory intervention often cuts across various different regulators, crypto markets operate at a global level within a regulatory void, and no globally harmonized position has been reached

²² Trautman (n 17) 3.

²³ De Nederlandsche Bank, AFM, *Cryptos - Recommendations for a Regulatory Framework*, (December 2019), p. 16-17.

<<https://www.afm.nl/en/nieuws/2019/jan/adviesrapport-crypto>> accessed 20 August 2020.

²⁴ Chung Baek, Matt Elbeck, 'Bitcoins as an Investment or Speculative Vehicle? A First Look' (2014) 20 App Econ L 1 30.

²⁵ Raphael Auer, Stijn Claessens, 'Regulating cryptocurrencies: assessing market reactions' (2018) BIS Quart Rev 51; Primavera de Filippi, 'Bitcoin: a regulatory nightmare to a libertarian dream' (2014) 3 Int Pol Rev 2 43.

yet. The fragmented approach with varying degrees of national regulatory scrutiny may lead to the race to the bottom, as crypto-asset related activities might migrate towards less stringent regulatory regimes. In addition to the EU institutions as representatives of regional integrations, several standard-setting international organizations are involved in considering the risks of using cryptocurrency and reviewing their regulatory approaches. The Committee on Payments and Market Infrastructures of the Bank for International Settlements (BIS) focuses on central bank digital currencies, the International Organization of Securities Commissions on ICOs and investment crypto-assets, the Basel Committee on Banking Supervision (BCBS) on banks' exposure to crypto-assets, the Financial Action Task Force (FATF) on anti-money laundering and anti-terrorism financing measures. Regarding the scope of the adopted standards in national legislations, the role of the FATF is most emphasized, which will be shortly presented in this paper, since it has grown into a transnational regime and most countries strive to harmonize national legislation with the FATF recommendations. Fearing hard regulation and a command-and-control regulatory approach, self-regulatory organizations, created to represent the interests of the fintech industry, are increasingly engaged in promoting self-regulatory rules. Self-regulatory organizations aim to formulate a common set of guidelines and professional standards of conduct to promote the integrity, fairness, and efficiency of markets in crypto-assets. One of the best examples is the Code of Conduct developed by the Association for Digital Asset Markets (ADAM).²⁶ Another example is the Virtual Commodity Association (VCA) that has recently announced a project to form a self-regulatory organization to issue guidance on best practices for crypto-assets, with the aim to form a basis for the future regulation of the sector and cooperation with existing regulatory bodies.²⁷ When the choice of forms of regulation and regulatory instruments is set as a criterion for the selection of the regulatory approach, somewhat modified classification proposed by the Committee on Payments and Market Infrastructure differentiates between the following: 1) information regulation and moral pressure; 2) regulation of specific financial intermediaries; 3) application of existing regulations; and 4) expansion of the scope of application of regulations, including the adoption of new regulations.²⁸

²⁶ <<http://www.theadam.io/code/>>

²⁷ <www.virtualcommodities.org>

²⁸ Bank for International Settlements, (n 16) 1; Gerard V. Comizio, 'Virtual Currencies: Growing Regulatory Framework and Challenges in the Emerging Fintech Ecosystem' (2017) 21 N. C. Banking Inst. 131, 168-170.

1) Moral pressure as a form of influence on the virtual assets market and information regulation are focused on the risks and dangers of investing in virtual assets and opening this market to the general public. Typical examples of this approach are public warnings about the risks of investing in cryptocurrencies, information campaigns on virtual currency schemes and the like. This form of regulation, with the least pressure on the markets, is often a precursor to the introduction of other forms.

2) Within the virtual currency system, some of the new intermediaries are providers of specific services that represent core elements of the market infrastructure. Examples of specific regulation of financial intermediaries are cryptocurrency exchanges and digital wallet service providers. Licensing systems for other intermediaries in cryptocurrency markets, which have similar roles to existing financial intermediaries, may impose a number of obligations such as minimum standards of investor protection (especially for consumers), risk reporting obligations and the like.

3) Cryptocurrency schemes may be regulated through the existing regulations, so that certain forms of intermediation and certain services are covered by already existing rules. This applies in particular to the rules on markets in financial instruments and regulations prohibiting market manipulation in the issuance and trading of investment tokens, as well as the application of rules on payment services to virtual currency schemes that have dominant payment system characteristics.

4) Expanding the scope of application of existing regulations and eventual adoption of new regulations is an option that seems necessary in situations when it is not possible to apply existing regulations due to the specific features of crypto-assets or the national regulators opt to harmonize rules with the transnational regime. For example, regulations on the prevention of money laundering and terrorist financing have been expanded to include cryptocurrency transactions as a result of the FATF recommendations. In addition to the need to include new service providers and new infrastructure (e.g. virtual wallets), digital assets are specific in relation to a number of other issues such as settlement finality, custody services, specific obligations towards customers, etc. One of the key dilemmas in defining the generic approach to crypto-asset regulation i.e. the need to adjust regulations or adopt a new regulation, refers to determining the nature of crypto-assets. That is why one of the leading principles in regulating crypto-assets rests on the notion of equal status. The principle may

be summarized as follows: "same services, same risks, same rules and same supervision". The application of this principle would lead towards an inclusive ecosystem where all actors are subject to similar rules, to ensure integrity of the financial system. Another principle derives from this attempt to constrain regulatory arbitrage. If the same risks attract the same rules, the regulation should be neutral regarding business models and technological neutrality. To harmonize regulatory standards on a risk basis, crypto-assets as financial instruments should be aligned with existing regulation, with considerations for a possible exemption regime. Fintech requires the regulatory framework to be based on the specific activity or function, rather than specific entity or the type of underlying technology. The principle of technological neutrality goes along with the principles-based regulation. Principles, as general requirements that express fundamental obligations that all market operators should observe, could be supported by more detailed standards. In general, principles-based regulation denotes a shift from reliance on detailed, prescriptive rules, towards more high-level, broadly formulated principles.²⁹ In a changing environment, regulatory approach should be resilient and adaptive. This implies that all new legislation and guidance and future amendments should provide for rapid changes and be adaptive to allow application to emerging technologies with no or limited amendment. Adopting a phased and unified approach based on risk is in line with this principle. Regulatory actions should be timeously assessed before stringent requirements are imposed. Safety, stability and integrity of the financial system is an underlying rationale for intervention and measures should commensurate with the level of risks, taking into account potential benefits. Unified approach means that all affected regulatory authorities should jointly determine their actions to ensure clear and consistent treatment of market participants.³⁰ The emergence of crypto-assets significantly contributed to disintermediation of traditional regulated intermediaries, such as commercial or investment banks. As middlemen, traditional financial intermediaries acted as gatekeepers to which semi-regulatory functions were attributed.³¹ Traditional

²⁹ Julia Black, Martin Hopper, Christa Band, 'Making a success of principles-based regulation' (2007) 3 Law and Fin. Markets Rev. 1 191.

³⁰ South African Intergovernmental Fintech Working Group (IFWG), Crypto-assets Regulatory Working Group, Position Paper on Crypto-Assets (2020) at p. 22-23. <http://www.treasury.gov.za/comm_media/press/2020/20200414%20IFWG%20Position%20Paper%20on%20Crypto%20Assets.pdf> accessed 15 September 2020.

³¹ For the instances of failure of gatekeepers in conducting their semi-regulatory functions, see Jennifer Payne, 'The Role of Gatekeepers' in Niamh Moloney, Eilís Ferran Jennifer Payne (eds) *The Oxford Handbook of Financial Regulation* (OUP 2015).

intermediaries were subject to direct and centralized regulation of their establishment and activities. The nature of open-sourced technologies and its constant evolution calls into question the possibility of applying a direct approach to regulation. Whereas potential externalities within crypto-market ecosystem are governed by the rule of the code, main risks may emerge in interactions with the real world. The emergence of various forms of new middlemen in the digital industry calls for a shift in the policy approach. Instead of regulating crypto technology, which is often not feasible due to its inherent characteristics, regulatory strategy could be implemented through the emerging middlemen and traditional intermediaries, such as banks, exchanges and payment institutions.

Traditional intermediaries were often regulated by regulatory measures focused on the industry structure, strategies or the activities performed by regulated entities. In regulating crypto-markets, direct regulation would be focused on the code or industry design and design features of the blockchain (i.e. node operators, miners), whereas indirect regulation would predominantly target interactions with counterparties and the real world.³²

These are, for instance, exchanges where cryptocurrencies are exchanged for a legal tender, merchant points, electronic wallet providers etc. A tool for direct regulation of crypto-assets may be the design-based regulation which aim is to remove the possibility of non-compliance by elimination of the possibility of human discretion.³³

However, internal self-regulation of the cyberspace and the code as architecture is a direct self-regulatory system. Direct regulation of the intersection of decentralized crypto world with the real world makes it a hard case for a centralized governance scheme. That is why a direct regulatory approach to crypto-markets would be a difficult endeavor, although it would not be completely ineffective. Centralized direct regulation of cryptocurrencies could face the same challenges that the traditional command-and-control regulation in an environment where the regulated industry is not sufficiently motivated to comply with such rules.

³² For the application of the concept of direct and indirect regulation in the regulation see: Hossein Nabilou, Alessio M. Paccos, 'The Hedge Fund Regulation Dilemma: Direct Vs. Indirect Regulation' (2015) 6 Wm & Mary Bus. L. Rev 183.

³³ Karen Yeung, 'Towards an Understanding of Regulation by Design' in Roger Brownsword and Karen Yeung (eds.) *Regulating Technologies: Legal Futures, Regulatory Frames and Technological Fixes*, (Portland, Oregon: Hart Publishing 2008), 80, 106.

From among all the challenges that centralized direct regulatory approaches face, three of them particularly stand out in regulating crypto-markets: limitations on the competence of public authorities, determination of the regulated entities, and regulatory arbitrage.³⁴ Since crypto-assets often encompass the traits of different products, the surveillance would cut across the traditional competences of various regulators, and traditional division on regulatory and supervisory tasks could be obsolete.³⁵ The lack of consensus on the nature of crypto-assets (*what* to regulate) prevents the emergence of a uniform approach to regulation. Consequently, another practical question emerges: *whom* to regulate? In a decentralized blockchain structure, an entity towards which the regulation could be directed could not be identified easily. There is a certain number of middlemen playing distinct roles, such as issuers of coins, miners, exchanges and various other actors.³⁶ Hence, it seems that indirect regulation is a more appropriate way, as it could be focused on the intersection of crypto world with the real world and target intermediaries such as exchanges and wallet providers.³⁷ Regulation could be deployed at the point where cryptocurrencies intersect with banks and payment institutions. Direct regulation, however, may face a number of challenges, such as regulatory arbitrage, which could only be mitigated through global regulatory coordination.³⁸

The decentralized nature of underlying technologies makes indirect regulation particularly convenient for a technology based "polycentric co-regulatory" regime.³⁹ Under this approach, financial supervisors would supervise the entities enabling the interface and interaction between crypto-assets and money, financial instruments or real assets. For instance, banks and payment institutions offering cryptocurrency accounts or investors buying ICOs may be subject to

³⁴ Hossein Nabilou, 'How to Regulate Bitcoin? Decentralized Regulation for a Decentralized Cryptocurrency', (2019) 27 Int J. on Law and Tech 3 266.

³⁵ Bank for International Settlements, 'Cryptocurrencies: Looking Beyond the Hype', *Annual Econ Report* (Basel, Switzerland 2018), 108.

³⁶ European Central Bank, *Virtual Currency Schemes - a Further Analysis* (Frankfurt 2015), 7-8,

<<https://www.ecb.europa.eu/pub/pdf/other/virtualcurrencyschemesen.pdf>> accessed 15 September 2020.

³⁷ Bank for International Settlements (n 35) 107.

³⁸ Ibid.

³⁹ Julia Black, 'Decentring Regulation: The Role of Regulation and Self Regulation in a 'Post Regulatory' World' (2001) 54 CLP 1 103.

prudential measures set in place to manage liquidity risks. Such an approach would presuppose adjustments in the corpus of prudential rules.

4.2. Information campaigns and moral suasion as alternatives to regulation

In pursuing regulation of innovative markets, regulators consider a ladder approach to regulation. First forms of intervention in cryptocurrency markets have been limited to moral suasion, user and investor-targeted information campaigns. In the initial stages of scrutinizing cryptocurrency markets, central banks were often the first authorities to issue warnings and papers in which they aimed to clarify the legal tender laws to explain that cryptocurrencies do not constitute legal tender.

National regulators then launched information campaigns as an alternative to information regulation. Such milder forms of intervention in the regulatory process were primarily focused on the risks and dangers of using and investing in cryptocurrency. Consumer-focused information campaigns usually give a brief overview of the concept of virtual currencies, and then dedicate the most space to identifying the risks that consumers should keep in mind when buying, using or trading virtual currencies. Warnings as forms of information campaigns may also contain recommendations on ways to protect or reduce the risks to which consumers are exposed.

One of the first examples of an information campaign is the Warning for Virtual Currency Users issued by the European Banking Agency in 2013, which presented several main risks to consumers.⁴⁰

Written in a clear language, the Warning explains what are the virtual currencies, how they function and what characteristics and risks should consumers be aware of when buying, holding, or trading virtual currencies, including the risk of being subject to tax liabilities. In countries where the use of cryptocurrencies of an investment nature has been developed, regulatory authorities issue special warnings about investments in the so-called initial public coin offerings (ICOs). For instance, the UK Financial Conduct Authority (FCA) published Consumer Warning about the Risks of ICOs in September 2017,⁴¹ and somewhat later a Guidance on Cryptoassets.⁴²

⁴⁰ European Banking Authority, *Warning to Consumers on Virtual Currencies*, EBA/WRG/2013/01, (2013).

⁴¹ Financial Conduct Authority, *Consumer Warning about the Risks of Initial Coin Offerings ('ICOs')*, (2017),

<<https://www.fca.org.uk/news/statements/initial-coin-offerings>> accessed 20 October 2020.

In the comparative regulatory practice of information campaigns, a difference can be noticed between countries whose campaigns were focused on warning consumers to refrain from investing in cryptocurrencies and not to use them as a method of payment for goods or services and those which focus on explaining core risks. Supervisors often warn consumers that rules on regulated financial products which functionally correspond to the types of virtual assets are not applicable. To conclude, a similar feature of all campaigns is the warning against potential risk and recommendation to consumers to be cautious when buying and trading in crypto-assets. On the other side, the publication of research papers and other informative studies to raise awareness of the risk of crypto-currencies may also be considered as an informational approach and an alternative to regulation.

4.3. Rules vs Principles and Regulator's Guidance

To increase the efficiency of regulation, regulators may consider a following step in the ladder approach to regulating crypto-markets. In direct or indirect form of regulation, regulators could restrict the issuance of crypto-assets, focus on market capitalization of assets and regulate the ones that pass certain thresholds in terms of size. In its extreme form, the regulator could ban the issuance and trading in crypto-assets and impose sanctions on crypto-based financial activities.

In a general way, the approach to crypto-assets may be classified within the restrictive or permissive model. The restrictive model, as its name suggests, is where regulation prohibits some or all the activities relating to virtual assets. It is likely to stifle innovation, prevent the development of good market standards, drive activity underground and thus increase the risk of criminal activities related to virtual assets. This model should clearly delineate what is permitted and what is not, to encourage business to understand the scope of restrictions. The permissive model sets out, by way of hard or soft law, the scope of permitted activities. It could be based around overarching principles or mandated outcomes, which must be achieved in order for the activity to be permitted. Enforcement is a challenge to this approach, particularly due to the extra-territorial character of crypto markets. To avoid confusion on the application of principles and achieving outcomes, and to deter businesses from operating in gray areas, rules and standards should be formulated in a clear way and interaction with existing laws should be achieved to decrease the risk of

⁴² Financial Conduct Authority, *Guidance on Cryptoassets*, Consultation Paper 19/3, (January 2019). <<https://www.fca.org.uk/publication/consultation/cp19-03.pdf>> accessed 20 October 2020.

enforcement. When based on principles and guidance, the permissive regime ensures flexibility to participants as to how they would comply.

When hard law is taken as a criterion, primarily in the context of the so called command-control regulation, we can distinguish between regimes of complete or partial prohibition, as well as partial and complete regulation. For some authors, the establishment of separate legal framework applicable only to crypto-asset activities is a "bespoke regulation", whereas under "bespoke regulatory regime" they refer to a distinct regulatory framework applied to a set of a wider fintech activities or DLT.⁴³

Full or partial ban on activities imposed through hard law is based on consideration that the risky crypto market should be banned, due to its decentralized nature and the fact that the authorities cannot block internet access. However, a total ban on the use of virtual currencies would probably fail in practice or drive activities underground. Some countries have explicitly banned cryptocurrencies, i.e. Bitcoin: Bangladesh, Nepal, Kyrgyzstan, Bolivia, Ecuador, Indonesia and Algeria. For example, Algeria passed a Law in late 2017 banning the "purchase, sale, use and holding of so-called virtual currencies" within the country.⁴⁴ China banned ICOs and crypto-asset exchanges and blocked access to all domestic and foreign cryptocurrency exchanges in 2017, and subsequently 2018 China also prohibited financial institutions from handling cryptocurrency transactions and ordered banks and payment institutions to close Bitcoin trading accounts. Partial prohibition, viewed from another standpoint, could be presented as partial or selective regulation. In order to reduce the impact of cryptocurrencies on the real economy, regulators in some countries prohibit certain ways of using cryptocurrencies. For example, the exchange of products and services could be prohibited, or the establishment of cryptocurrency exchanges based in the country could be forbidden.

These include bans aimed at preventing the use of cryptocurrencies for illegal activities (money laundering), which is the most common form of the partial prohibition. Partial prohibitions are, as a rule, instruments of direct regulation.

⁴³ Appoline Blandin et al., *Global Cryptoasset Regulatory Landscape Study*, Cambridge Centre for Alternative Finance (2019)

<<https://www.jbs.cam.ac.uk/wp-content/uploads/2020/08/2019-04-ccaf-global-cryptoasset-regulatory-landscape-study.pdf>> accessed 25 October 2020.

⁴⁴ Art. 117, Loi N° 17-11 du 8 Rabie Ethani 1439 Correspondant au 27 Decembre 2017 portant Loin de Finances pour 2018; <https://www.mfdgi.gov.dz/images/pdf/lois_de_finances/LF2018F.pdf> accessed 15 August 2020.

Additional grounds for indirect regulation are, for instance, when regulator focuses on regulating liquidity providers in cryptocurrencies, and impose stricter collateral eligibility requirements on the acceptance of cryptocurrencies as collateral by financial intermediaries.⁴⁵ Higher prudential requirements could apply to offset the higher risks of cryptocurrencies. Qualitative and quantitative prudential rules on credit or payment institutions in their relations with participants in crypto-markets is an alternative to ring-fencing the traditional banking and payment systems, which could be considered as a policy option.

Partial regulation implies the adoption of regulations regarding cryptocurrencies which enable their regulated use and clear treatment in certain aspects, important for the state. One example could be to prescribe tax treatment of using cryptocurrencies. Comparative legal analysis reveals that most countries are moving towards a moderate approach, where, as a rule, more detailed transactions within cryptocurrency markets are not regulated (e.g. exchange of one cryptocurrency for another), and attention is focused on preventing the use of virtual currencies to financial stability or harm public interest. In this sense, partial regulation can be identified with a partial prohibition (e.g. a special regime to prevent the use of cryptocurrencies as a measure to prevent money laundering and terrorist financing).

Detailed regulation of cryptocurrencies can be focused on participants, the services they provide, as well as possible risks. Organizers of cryptocurrency schemes and service providers are given the status of financial intermediaries (usually through various licensing schemes). More or less supervised, they are assigned various obligations by which the regulatory authorities try to uncover the risks of using cryptocurrencies.

Some regulatory systems are mostly focused on the so-called risk hubs, such as digital wallet providers and cryptocurrency exchanges.⁴⁶ Finally, the algorithms themselves used in fintech innovations could be regulated in order to control their correctness and transparency.

The scope of a permissive and more detailed form of regulation of crypto-market intermediaries, services and infrastructures, depends on the typology of crypto-assets as a diffuse concept. As will be explained further in this Paper, the principle "same activity, same regulatory framework", is being applied by an increasing number of regulators.

⁴⁵ Bank for International Settlements (n 35), 108.

⁴⁶ International Monetary Fund, *Fintech and Financial Services: Initial Considerations*, IMF Staff Discussion Note 17/05, (Washington 2017).

It indicates that economic function of one crypto-assets must be compared to the other forms of crypto-assets, or other financial products. Therefore, the starting point in defining the rules and regulating the market should be their dominant economic function (payment, investment, realization of some benefits), as well as similarities and differences compared to other forms of real and financial assets.

Due to their evolving nature and complexity, indirect regulation of crypto-assets appears to resemble more closely standards, as opposed to rules.⁴⁷ The indirect approach to regulation shifts the focus from rules based regulation into principles based regulation, when intermediaries who transact with providers of crypto infrastructure or services implement it. This is due to the fact that such intermediaries, as surrogate regulators, will enforce rules in a more decentralized way, with more flexibility in implementation, which is conditioned by the very nature of the fintech that is constantly evolving.⁴⁸ However, overreliance solely on rules or only on principles is a "black or white" approach, as principles serve to retain a greater degree of congruence in comparison with static prescriptive rules. It is important to acknowledge that fintech regulation should be a set of complex norms containing both rules and principles, as the majority of complex risk regimes in practice contain a mixture of both rules and principles.⁴⁹

If the primary goal of the regulatory strategy is to encourage innovation as well as resource savings, the guidance model allows for a more flexible approach. However, this regulatory model faces one key problem: its effective implementation. Excessive generalization of principles and their vagueness may deter operators from innovating in areas that are not regulated, or motivate them to circumvent rules relating to services that have a similar economic function and are subject to detailed regulation (e.g. regulations on electronic money, payment services, securities market, etc.).

Within this model, guidelines are determined either by laws and bylaws, or by interpretative acts of administrative bodies that, as a rule, do not represent a formal source of law. Regulatory guidance has predominantly been focused on applicability of securities laws, and to a lesser extent rules on electronic money and payment services. The guidelines determine the scope of activities related to

⁴⁷ On this issue see, for example, Cass. R. Sunstein, 'Problems with Rules' (1995) 83 Cal. L. Rev. 4 1021; Luis Kaplow, 'Rules Versus Standards: An Economic Analysis' (1992) 42 Duke L. Jour. 2 557.

⁴⁸ Hossein Nabilou, Alessio M. Paces (n 32).

⁴⁹ Christie L. Ford, 'New Governance, Compliance, and Principles-Based Securities Regulation' (2008) 45 Am. Bus. L. J. 1, 10; Julia Black et al. (n 29) 193.

virtual assets and set requirements in the form of principles and results, that must be achieved by their application in order for a certain activity to be performed. Gibraltar is a good example of a broad-based model based on guidelines, that encompasses the entire field of distributed ledger technologies.

The license of the Financial Services Commission of Gibraltar is required for all activities based on DLTs, if used to store or transfer value which belongs to third parties. However, the coverage is set more restrictively, given that qualified intermediaries cannot be entities that intend to use virtual currencies to pay for goods and services, nor financial institutions that have already been licensed to operate in Gibraltar.

The broad nine principles, each of which is accompanied by a more detailed instruction, leave the supervisory authority more flexibility in deciding whether to grant a license and revoke it.⁵⁰ However, it is possible for guidelines to regulate only one activity or service provision within virtual currency schemes that have a specific economic function. As mentioned already, the UK Financial Conduct Authority - FCA has published a Guidance on Cryptoassets.⁵¹

The FCA formulated recommendations to financial intermediaries regarding the risk of application of existing regulations in the field of capital markets that refer to regulated financial investments and instruments. The Guidance, in the same time, represents a guideline for determining the so-called regulatory perimeter, which represents the boundary separating regulated from unregulated activities.

All activities falling within the FCA regulatory perimeter domain require the permission of this body or the Prudential Regulatory Authority, which is issued in accordance with the Law on Financial Services and Markets and the accompanying Regulation.⁵²

The significance of this Guidance is to provide clarification, with practical examples, in which situations payment tokens, investment tokens and utility tokens will be subject to existing regulations or be outside the competence of regulatory authorities.

⁵⁰ Gibraltar Financial Services Commission, Distributed Ledger Technology Providers - Guidance Notes,

<<https://www.gfsc.gi/downloads?section=19&type=0>> accessed 20 October 2020

⁵¹ Financial Conduct Authority, Guidance on Cryptoassets, Consultation Paper 19/3, January 2019. <<https://www.fca.org.uk/publication/consultation/cp19-03.pdf>> accessed 20 October 2020

⁵² <<http://www.legislation.gov.uk/ukxi/2001/544/part/II/made>> accessed 20 October 2020

4.4. Measures to promote innovations in the Regulatory Process

New technologies have introduced a new chapter in economic history and represent an example of how advances in technology pose challenges to regulators and supervisors. Many regulators are procrastinating with regulations to observe the potential benefits of DLT technology. Not to be in touch with the fintech revolution, or following a precautionary approach is risky.

For instance, prohibiting financial engineering, before having observed its functioning, could heavily impede innovation in the financial sector, lose fintech companies to other jurisdictions, and ultimately slow down the economic growth.⁵³

Regulators who keep a track in fintech face a challenge on how to design a flexible regulatory environment to accommodate fundamental changes to the financial market and foster competition in financial services. Such regulators understand that high market barriers in the financial sector and a lack of competitive pressure are deterrent to development of financial intermediation, and therefore have an increased focus on competition and innovation in financial system. In order to make new entrants more attractive, governments and financial regulators are developing and/or sponsoring innovation facilitatory units dedicated to foster financial innovation. Innovation facilitators emerged as an alternative and experimentalist governance model to promote technological innovations. Innovation hubs operate as a prompt information desk providing a non-binding guidance, clarification and support regarding regulator's expectations.

On the other hand, it enables regulators to monitor market integrity and investor protection, and provides relevant data on innovations and potential risks.⁵⁴ In most countries, innovation hubs are established within financial regulators or in a networked mode of cooperation among financial regulators and other authorities. Malta is an example where the legislative package of 2018 resulted in setting up the Digital Innovation Authority, which closely cooperates with Malta Financial Services Authority.⁵⁵

⁵³ John Armour et al., *Principles of Financial Regulation* (OUP 2016), 51.

⁵⁴ Ivana Bajakić, 'Transformation of financial regulatory governance through innovation facilitators - case study of Innovation Hub in Croatian Capital Markets', (2020) 4 *EU and comparative Law Issues and Series (ECLIC)*, 917, 922-923.

⁵⁵ The Virtual Financial Assets Act, Chapter 590 of the Laws of Malta (the VFA Act); the Innovative Technology Arrangements and Services Act, Cap 592 of the Laws of Malta (ITASA); the Malta Digital Innovation Authority Act, Cap 591 of the Laws of Malta (the MDIA Act).

Many modern regulators aim to waive unnecessary regulatory burdens and facilitate an enabling environment for mutual learning. The new regulatory approach allows regulator to foster a dialogue between fintech operators and better understand new technologies, which supposedly provides the flexibility to react to unknown risks. This institutionalized dialogue is a controlled space in which fintech firms may test and validate innovative products, infrastructure and services, for a limited time, and where regulators provide interpretations and help them to comply, is known as "regulatory sandbox".⁵⁶

Although there are differences and variations between them in different countries, sandboxes share common policy objectives with regards to ensuring consumer and investor protection, market integrity and promoting innovation and competition.⁵⁷ Each regulator may define requirements under which firm are granted access and safety requirements that should be met by the potential participant. Testing periods are as a rule limited typically from 6 to 12 months, set on a case-by-case basis. Most differences among national approaches are related to the level of the compromise and the extent of the relief of regulatory requirements for firms. In any way, the sandbox reduces regulatory uncertainty and encourage fintech firms to experiment within a grey zone.

The main benefit of this approach is to prevent premature regulatory actions and to accelerate the assessment of risks of new technologies. However, the importance of regulators learning based on observance is limited, as some risks may emerge as the test time lapses. Sandbox approach may be resource-intensive and presupposes experienced staff in well-equipped and financially sound regulators. Liberal approach of those regulators who do not have expertise and resources may lead them to undertake unacceptable risks.⁵⁸

Therefore, in countries where regulators do not want to hinder the development of financial innovations, but aim to actively monitor the effects of new technologies, a restricted environment called a "regulatory sandbox" has been

⁵⁶ European Banking Authority (EBA), Discussion Paper on EBA's approach to financial technology (FinTech), (EBA/DP/2017/02, 4 August 2017), <[eba.europa.eu/sites/default/documents/files/documents/10180/1919160/7a1b9cda-10ad-4315-91ce-d798230ebd84/EBADiscussionPaperonFintech\(EBA-DP-2017-02\).pdf?retry=1](http://eba.europa.eu/sites/default/documents/files/documents/10180/1919160/7a1b9cda-10ad-4315-91ce-d798230ebd84/EBADiscussionPaperonFintech(EBA-DP-2017-02).pdf?retry=1)> accessed 15 October 2020.

⁵⁷ Financial Stability Board, Financial Stability Implications from Fintech: Supervisory and Regulatory Issues that Merit Authorities' Attention (27 June 2017) 4f <www.fsb.org/wp-content/uploads/R270617.pdf> accessed 10 September 2020.

⁵⁸ Dirk A. Zetsche et al., 'Regulating a Revolution: From Regulatory Sandboxes to Smart Regulation' (2017) 23 Fordham J Corp & Fin. L. 1 31, 79.

put in place. Up to date, regulatory sandboxes exist in around 40 countries.⁵⁹ Such an approach allows companies starting a business based on blockchain technology to operate in a relatively favorable and controlled environment at a given time, in order to test the advantages and disadvantages of applying new technologies. This achieves that participants in the development of new technologies are not too limited in a situation when the risks of using new technologies are not clear enough. On the other hand, they are given the opportunity to continue with financial innovations. This approach could be a form of supervised self-regulation of markets for innovative financial products.

5. BASIC ISSUES IN DETERMINING THE FOCUS OF REGULATORY INTERVENTION

5.1. *Ratione materiae*: functional approach to crypto-assets

The starting point in regulating crypto-assets is to determine the scope of application of the regulatory regime. The first challenge that regulators face is to be aware of the host of legal consequences in determination of what should be regulated.⁶⁰ Regulators and supervisors should be cautious in legal categorization of cryptocurrencies. With this in mind, the regulatory authorities as a rule take a functional approach and scrutinize the economic activities performed by digital assets. This may be complicated due to the fact that crypto assets may perform certain functions similar to those of fiat currencies, financial instruments and commodities, and may be subject to different activities and services.⁶¹ That is why an all-encompassing term "crypto-assets" or digital assets is best to encapsulate different crypto-asset forms under one umbrella.

In short, the creation of cryptocurrencies is one of the most important applications of the blockchain technologies in finance. As the value of crypto-assets primarily depends on cryptography and DLT, crypto-asset is considered to be a digital mean based on cryptographic functions which is recorded in the

⁵⁹ An updated list of existing sandboxes is compiled by the DFS Observatory, "Regulatory Sandboxes" <dfsobservatory.com/content/regulatory-sandboxes> accessed 10 October 2020.

⁶⁰ Rosa Maria Lastra and Jason Grant Allen, "Virtual Currencies in the Eurosystem: Challenges Ahead," (Brussels, Belgium: ECON Committee, European Parliament, 2018), 9.

⁶¹ For an overview of crypto-asset activities see: Global Cryptoasset Regulatory Landscape Study, the Cambridge Centre for Alternative Finance, 2019, 23-26, <https://www.jbs.cam.ac.uk/fileadmin/user_upload/research/centres/alternative-finance/downloads/2019-04-ccaf-global-cryptoasset-regulatory-landscape-study.pdf> accessed 15 September 2020

distributed ledger.⁶² The classification of crypto-assets is contested. The complex issue of the legal nature of crypto-assets, which no previous *acquis* can clearly classify within a taxonomy defined by the Law of property, has been the subject of numerous publications and scientific papers published in the past few years. It seems that it has become clear that it is not possible to apply the existing concepts, but it was necessary to create a new concept of digital assets.⁶³

Depending on the specific characteristics and functions of its forms, as well as the types of services offered by intermediaries in the crypto market, crypto-assets may be divided into three groups according to their economic function. The first group represents payment tokens or cryptocurrencies a means of exchange or payment. Some regulators use the term exchange tokens.⁶⁴ The second broad group of cryptocurrencies includes investment tokens or security tokens, and the third group represents utility tokens which holders are entitled to access specific products or services. Stablecoins may also be included as a specific category of more secure forms of digital property. The distinction between exchange tokens and utility tokens is not clear-cut, as a group of hybrid tokens combine elements of investment tokens as financial instruments and various benefits and advantages that do not have the character of a financial investment. Finally, the use of blockchain technologies and distributed records has led to the creation of platforms which record any data on physical or financial assets, such as precious metals, agricultural products, financial assets (stocks, bonds), etc. The process of "tokenization", the goal of which is to speed up the settlement of transactions and eliminate transaction costs, has urged the development of new forms of tokens called asset-backed tokens, i.e. tokens secured by property.⁶⁵ The following graph illustrates the distinction based on the economic function of crypto-assets.

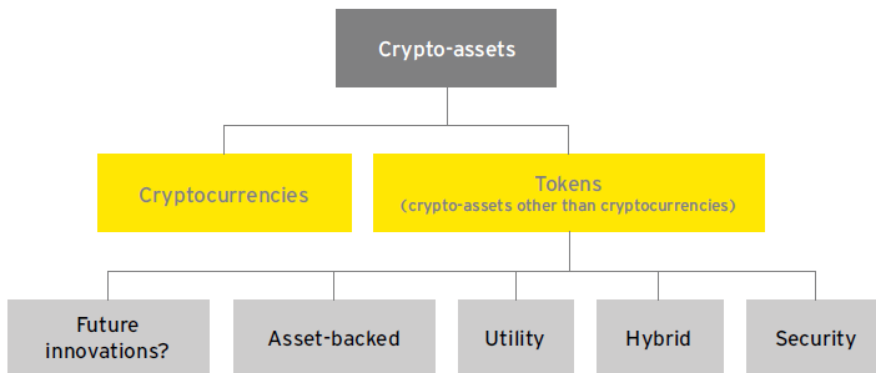
⁶² European Banking Authority, Report with advice for the European Commission on "crypto-assets" (January 2019).

⁶³ Shawn Bayern, 'Dynamic common law and technological change: the classification of Bitcoin' (2014) 71 *Wash. & Lee Rev Online*, 22.

⁶⁴ HM Treasury, FCA, Bank of England, *Cryptoassets Taskforce: Final Report* (October 2018),

<https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/752070/cryptoassets_taskforce_final_report_final_web.pdf> accessed 10 October 2020.

⁶⁵ A broad concept of asset-based tokens is adopted by the Swiss regulator, as both tokenization of debt and securities instruments are encompassed, Swiss Financial Market Supervisory Authority, *Guidelines for enquiries regarding the regulatory framework for initial coin offerings (ICOs)*, (February 2018)



Source: Ernst & Yeung, *Life of a coin: Shaping the future of crypto-asset capital markets*, 2019, at p. 9; <[https://www.ey.com/Publication/vwLUAssets/ey-life-of-a-coin/\\$File/ey-life-of-a-coin.pdf](https://www.ey.com/Publication/vwLUAssets/ey-life-of-a-coin/$File/ey-life-of-a-coin.pdf)> accessed 10 August 2020

The above taxonomy, based on the economic function, roughly encapsulates similar risks that arise due to their use. Albeit not entirely objective, it allows to determine the legal qualification of virtual assets based on blockchain technologies and DLT. At EU level, most of these forms and associated services are not subject to a specific regulatory regime and it is not entirely clear which rules of the broad acquis of European financial regulation could be applied.⁶⁶

The need to harmonize regulations with the requirements of modern technologies was recognized in the Action Plan for Financial Technologies,⁶⁷ which mandated European agencies to assess the applicability of the existing regulatory regime of financial services to cryptocurrency.

The advisory reports emphasized that some provisions of European legislation may negatively affect financial engineering, but also that in most cases regulations on consumer and investor protection are not applicable to transactions related to certain types of virtual assets.⁶⁸

<<https://www.finma.ch/en/news/2018/02/20180216-mm-ico-wegleitung/>> accessed 10 October 2020.

⁶⁶ Phillip Hacker, Chris Thomale, 'Crypto-Securities Regulation: ICOs, Token Sales and Cryptocurrencies under EU Financial Law' (2018) 15 Eur. Co. Finan. Law Rev. 645.

⁶⁷ European Commission, Communication: *FinTech Action Plan: For a more competitive and innovative European financial sector* (March 2018), <https://ec.europa.eu/info/publications/180308-action-plan-fintech_en>

⁶⁸ European Securities and Markets Authority, European Securities and Markets Authority, *Advice on Initial Coin Offerings and Crypto-Assets*, (January 2019), Reference No. ESMA50-157-1391;

Proposal for a Regulation of the European Parliament and of the Council on Markets in Crypto-assets (MiCA), and amending Directive (EU) 2019/1937 on the protection of persons who report breaches of Union law⁶⁹ has chosen to provide a baseline regulation for the general regime of crypto-assets (catch-all category of crypto assets) and exceptional regime for the specific crypto-asset types (asset-referenced tokens and e-money tokens). The increased specificity enters into the picture with asset-referenced tokens, electronic money tokens and utility tokens, which are all types of crypto-assets categorized under the regulatory framework of the Proposal. The Proposal has foreseen an important subclass of asset referenced tokens (significant asset-referenced tokens), as well as significant e-money tokens.

When defining cryptocurrency, special attention should be paid to the following two aspects, which necessarily affect the legal qualification and application of the existing rules applicable to financial markets. The first aspect refers to the coverage of financial instruments regulated by capital market regulations, which implies that some types of investment tokens may represent regulated financial instruments. Another aspect refers to the regime of electronic money and payment services, since some forms of cryptocurrency have great similarities with electronic money, and some payment systems based on blockchain may represent quasi-payment institutions.

From the point of view of the enforcement of existing regulations, one group of crypto-assets is especially important as it enables one or more rights in digital form to be stored, deposited or transferred through DLT, with the possibility of directly or indirectly identifying the property owner. It is created through so called "initial coin offering procedure" (ICO). From the point of view of comparative law and practice, many regulatory bodies are trying to extend the application of regulations regulating public offer financial instruments to these forms of crypto-assets. In its advisory report from January 2019, the European Securities and Capital Markets Agency pointed out that the key determinant of the legal status of virtual assets is the qualification of tokens as a financial instrument, in the sense of the Markets in Financial Instruments Directive (MiFID II).⁷⁰ Financial instruments are defined in Art. 4 (1) (15) of this

<https://www.esma.europa.eu/sites/default/files/library/esma50-157-1391_crypto_advice.pdf>

⁶⁹ Proposal for a Regulation of the European Parliament and of the Council on Markets in Crypto-assets, and amending Directive (EU) 2019/1937.

⁷⁰ European Securities and Markets Authority (n 68).

Directive and listed in more detail in Annex I. The scope of financial instruments set out in MiFID conditioned the application of a number of other regulations, such as the Market Abuse Regulation⁷¹ and the Prospectus Regulation.⁷²

Receiving and sending cryptocurrencies within virtual currency schemes of bidirectional flow is not considered a payment service in terms of regulations on payment services. As a note, according to Directive 2015/2366 on payment services in the internal market, cash means banknotes and coins, demand deposits and electronic money within the meaning of Directive 2009/110/EC on the supervision of electronic money institutions.⁷³

Issuers of virtual currencies are not considered issuers of electronic money.

The key challenge in defining the object of regulatory intervention is to shift from a traditional "entity based" approach, as different entities may carry out similar types of activities and hence similar risks, which may not be subject to the same licensing and conduct of business regulation. Activity based approach, on the other side, is technology neutral and focuses on similar services.

This approach has another benefit: it helps regulators not to overlap their responsibilities. Coordination and determination of competencies is primarily based on the economic function of the token, usually determined on the basis of financial instrument tests. These tests aim to determine the so-called "financial perimeter", or the extent of applicability of existing laws. The regulatory perimeter refers to the types of financial services activity to which the existing regulation is applied. Financial instrument test favor substance over form. As crypto-asset services may fall within the regulatory perimeter of several regulators who have mandates to regulate financial market and monitor financial stability, regulators facing fintech innovations need to coordinate their activities.⁷⁴

⁷¹ Regulation (EU) 596/2014 of the European Parliament and of the Council of 16 April 2014 on market abuse.

⁷² Regulation (EU) 2017/1129 of the European Parliament and of the Council of 14 June 2017 on the prospectus to be published when securities are offered to the public or admitted to trading on a regulated market.

⁷³ Directive 2009/110/EC of the European Parliament and of the Council of 16 September 2009 on the taking up, pursuit and prudential supervision of the business of electronic money institutions amending Directives 2005/60/EC and 2006/48/EC and repealing Directive 2000/46/EC.

⁷⁴ An example of regulatory coordination established to avoid the overlap of activities is the UK Cryptoasset Taskforce (n 64).

5.2. Licensing regime: establishing requirements for licensing institutions and activities

The model of licensing, as approval of performing a certain activity by a supervisory body (usually the one that regulates financial services), usually involves registration or issuance of a permit to operate, upon meeting prescribed requirements and the required criteria. Certain activities carried out in the country of origin of the participants in the crypto-asset markets, or in the country in which the transactions take place, are subject to a licensing or authorization regime, and the duty to reporting to supervisors. Licensing coverage varies from country to country, but intensive activities on changes to the legal framework and the adoption of special regulations on crypto-assets in recent years share some common characteristics.

One of the pioneers of virtual currency licensing schemes is the State of New York, whose New York Department of Financial Services launched a BitLicense regulatory framework in 2015, applicable to THE activities related to virtual currencies that take place in New York or are offered to its residents.⁷⁵

Providers of services related to virtual currencies offered to resident citizens and legal entities with registered office, or those who undertake related business activities in the territory of the State of New York, are obliged to apply for a license, without de minimis exceptions.

The licensing procedure does not differ significantly from the licensing procedure for financial institutions. Such an approach by New York State authorities has resulted in a relatively small number of licenses issued, while some virtual currency service providers explicitly refused to provide services to New York residents in order to avoid licensing requirements.⁷⁶

This law inspired the US Uniform Law Commission to propose a model law on the regulation of virtual currency transactions (the Uniform Regulation of Virtual-Currency Business Act) in 2017, which has so far been adopted in three US federal states.⁷⁷ The New York licensing requirements are applicable, in particular, to services related to the sale, exchange, deposit of funds in virtual

⁷⁵ Virtual Currency Regulation, 23 NYCRR Part 200, New York Financial Services Law.

<https://www.dfs.ny.gov/apps_and_licensing/virtual_currency_businesses/gn/adoption_licensing_vc> accessed 20 August 2020.

⁷⁶ Gerard Comizio (n 28).

⁷⁷ Uniform Law Commission, Regulation of Virtual Currency Businesses Act, <<http://www.uniformlaws.org/Act.aspx?title=Regulation%20of%20Virtual-Currency%20Businesses%20Act>>

wallets, storage of keys necessary for transactions, control, management or issuance of virtual currencies.⁷⁸

A license may be required in a detailed and partial regulation mode. Depending on the extent to which certain services and financial intermediaries are regulated within virtual currency schemes, a distinction could be drawn between pure and hybrid licensing models. The hybrid model is basically a model based on guidelines and often relies on principle-based regulation, because it involves registration with the supervisory authority, which has a more flexible approach and monitors whether the regulated entity adheres to the set principles. Even when the term "registration" is used within a certain national regulatory regime (including the absence of a discretionary assessment of the regulatory body), the registration process imposes a number of obligations on participants in crypto markets, such as the establishment of the internal control systems, identification of clients and implementation of procedures provided by regulations on prevention of money laundering, etc.

It is possible that a license is required for only certain activities or services, or in respect of specific financial intermediaries. One of the most common examples of such narrower licensing models is the requirement to license virtual currency exchanges, as is the case e.g. in Australia, Japan, the Philippines, etc. Registration by the financial regulator in these countries primarily refers to the operations of exchanges, while other activities are not regulated or may be subject to the application of general capital market regulations (e.g. issuance of investment tokens).

The main purpose of virtual currencies is to be means of payment. Payment institutions are increasing their involvement in the cryptocurrency business, whereas many cryptocurrency schemes have similar features and aim to become parallel payment infrastructure or alternative payment systems.⁷⁹ Licensing requirement is one of the first steps to bring virtual currencies within the regulated ecosystem, whether it is a specific license for activities on crypto-markets (e.g. BitLicense) or payment institution license for virtual currency exchanges. However, despite many benefits, such decentralized payment systems have unstable values, are subject to new operational risks, could be used

⁷⁸ N.Y. Comp. Codes R. & Regs. tit. 23, § 200.3 (2019)., New York Department of Financial Services, New York Codes, Rules and Regulations; Title 23, Department of Financial Services, Chapter I. Regulations of the Superintendent of Financial Services, Part 200, Virtual Currencies

⁷⁹ European Central Bank (n 36).

for illicit transfers and ultimately could jeopardize financial stability. Licensing crypto exchanges as payment institutions rises a number of risks, therefore directing the regulations towards the relationship of payment institutions and crypto-assets (alternative indirect regulation) could be a better strategy. A major setting where cryptocurrencies could be regulated is through prudential regime over regulated institutions which engage in the provision of payment services.⁸⁰

The licensing model is basically often based on the method of partial or detailed regulation. The subject of normative regulation could be intermediaries and infrastructures; types of digital assets and activities or services related to crypto-assets; relations, especially relations with clients who represent non-professional investors (consumers), as well as public order requirements such as the issue of taxation and prevention of money laundering and the use of virtual currency schemes for illegal purposes. Crypto-asset regulation does not, as a rule, cover end-users of virtual currencies, such as traders who accept them when paying for goods or services, consumers who buy goods or services, as well as investors who invest in virtual currencies. The subject of regulation are primarily activities that are reflected in the transfer of virtual currencies and services related to transactions. Indirect regulation is considered to be a more appropriate way, as it could be focused on the intersection of crypto world with the real world and target intermediaries such are exchanges and wallet providers.⁸¹ Targeting the interface of cyberspace and material world is in line with the old tradition in financial regulation which relied on regulated gatekeepers.⁸²

As was the case with classifying crypto-assets, regulators struggle how to refer to different types of crypto-assets activities: creation, initial distribution and secondary trading. In relation to these activities, a variety of different infrastructures and services emerged, among which some perform tasks specific to the nature of crypto-assets (i.e. miners), while others have roles which exist in traditional financial services (i.e. exchanges). New activities are specifically enabled by the properties of crypto-assets and the underlying infrastructure, whereas mixed activities combine characteristics of both traditional and new activities. The majority of crypto-asset related activities carried out by intermediaries are quite similar to traditional activities (such as exchange and

⁸⁰ Hossein Nabilou, "The Dark Side of Licensing Cryptocurrency Exchanges as Payment Institutions," *SSRN Working Paper Series* (2019), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3346035 accessed 07 November 2020.

⁸¹ Bank for International Settlements, (n 35) 107.

⁸² Jennifer Payne (n 31).

trading platforms, derivatives, custody, payment services etc.) Some activities were adapted to better correspond to the new characteristics of specific crypto-assets. A relatively small number of activities specific to crypto-assets intermediation can be considered completely new and uniquely enabled through crypto-assets.

Licensing of intermediaries and institutions that form part of the infrastructure of the cryptocurrency industry is usually based on the general requirements provided for financial intermediaries in the comparative financial systems. When applying for a license (and often registration), stakeholders must meet the requirements of fit and proper standards, which apply to owners, or persons with special powers and responsibilities, as well as to establish an adequate system of corporate governance and internal controls. In some countries, there are requirements for the existence of a minimum amount of founding capital, i.e. the minimum capital that financial intermediaries and service providers related to virtual currencies must have in their accounts in order to ensure the protection of customers' funds. Requirements are increasingly being set for persons who have a qualifying or controlling interest, as well as for persons associated with institutions that provide services related to virtual currencies. In this sense, in some countries the requirements are closer to the requirements set before payment institutions, and in the case of virtual currencies of an investment nature may reflect the main traits of the investment companies' regulation.

Licensing may be provided as mandatory for certain entities and activities, and in other cases optionally, which is a form of incentive and promotes good business practice, while at the same time encouraging innovation. France is an example where all providers of the purchase and sale of virtual assets for money as legal tender, as well as providers of digital wallets and storage of virtual assets, are subject to registration by the Financial Markets Agency (AMF), primarily due to the requirements of anti-money laundering legislation.⁸³ In addition, the consent of the Prudential Control Agency is required, which controls the fulfillment of minimum capital requirements. Other intermediaries do not need a license, which if obtained represents a marketing instrument that allows them to differentiate in the market.

Given the numerous and unpredictable risks, intermediaries offering infrastructure and services in crypto-markets are also required to develop internal control and risk management systems, especially IT system risk as a type of operational risk. From the regulator's point of view, the most important

⁸³ L. 54-10-3 - L. 54-10-5 Code Monétaire et Financier.

reason for establishing internal control policies is to harmonize operations with the requirements set by regulations on the prevention of money laundering and terrorist financing. The EBA's "Opinion on Virtual Currencies" emphasizes the need for an agreement to be reached at EU level for participants dealing professionally with virtual currencies to be legal entities, and to be responsible for the integrity of the central ledger of transactions, protocols and other functional components of virtual currency schemes, as well as to meet all regulatory requirements through the functioning of the internal control system and adopt measures to protect individual users. The issue of internal controls is related to prudential requirements, as well as the regime of protection of public interest (anti-money laundering) and protection of users, which deserves to be addressed in the next title.

5.3. Core issues related to the protection of public economic order: measures to combat money laundering and customer protection

Cryptocurrencies have a number of features that are not yet covered by regulations, which means that there are numerous possibilities for abuse. The three key factors that influence this are the following: anonymity in the use of cryptocurrencies, their decentralized nature, and the lack of legal regulation in this area.⁸⁴ The biggest fears of violating the public economic order are the possibility of cross-border money laundering, legalization of property acquired through the commission of criminal acts, tax evasion, violation of regulations on foreign exchange operations and capital movements, and protection of consumer rights. A special issue is the operational risk of internet business, i.e. the sensitivity of virtual currency schemes and intermediaries to hacker attacks and various frauds.

There is a noticeable tendency of the emergence of transnational harmonized regime aimed at combating money laundering and terrorist financing, which is encouraged by the FATF recommendations. The amended Recommendation 15 requires jurisdictions to regulate crypto-assets and crypto-asset service providers for anti-money laundering, and to ensure that crypto-asset service providers are licensed or registered, and subject to effective systems for monitoring and supervision. FATF Guidance for a Risk Based approach represents an important policy document assisting countries and virtual asset service providers in understanding their anti-money laundering and counter-terrorist financing

⁸⁴ FATF/GAFI, *Virtual currencies – Key Definitions and Potential AML/CFT Risks*, 2014, <<https://www.fatf-gafi.org/media/fatf/documents/reports/Virtual-currency-key-definitions-and-potential-aml-cft-risks.pdf>> accessed 17 October 2020.

obligations, and implementing the FATF's requirements as they apply to virtual assets.⁸⁵

In its opinion of 2017, European Banking Agency⁸⁶ stressed that cryptocurrency exchanges and digital wallet service providers fall under the jurisdiction of the Fourth Anti-Money Laundering Directive,⁸⁷ which was an important step in limiting this risk. However, it was only with the adoption of the Fifth Money Laundering Directive⁸⁸ that the scope of application of the regulation was unequivocally extended to digital wallet service providers and virtual currency trading platforms.

Rules on crypto-assets adopted worldwide show that the regulation of crypto-asset-related activities exposes the main characteristics of regulating financial institutions. These include general principles of integrity, financial prudence, orderly market conduct, transparency, protection of clients' assets and avoidance of conflict of interest. Crypto-asset issuers and traders are subject to common fiduciary responsibilities, including in particular anti-money laundering provisions. Exchanges are increasingly obliged with the rules on orderly price discovery, segregation of client monies and formation and avoidance of front-running, collusion and in general market manipulation.

As a rule, the normative framework of virtual currency schemes does not include the relationship with end users, nor the relationship with traders who accept virtual currencies when paying for goods or services. The subject of regulation are primarily activities that are reflected in the transfer of virtual currencies and related services. One of the first examples of bespoke regulatory frameworks, which prescribed specific obligations to clients and consumer protection and specific procedures on client protection which are prerequisites for licensing, is

⁸⁵ FATF (n 2).

⁸⁶ <<https://www.eba.europa.eu/-/eba-publishes-an-opinion-on-the-commission-s-proposal-to-bring-virtual-currency-entities-in-the-scope-of-the-anti-money-laundering-directive> > accessed 20 October 2020.

⁸⁷ Directive (EU) 2015/849 of the European Parliament and of the Council of 20 May 2015 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing, amending Regulation (EU) No 648/2012 of the European Parliament and of the Council, and repealing Directive 2005/60/EC of the European Parliament and of the Council and Commission Directive 2006/70/EC, OJ L 141.

⁸⁸ Directive (EU) 2018/843 of the European Parliament and of the Council of 30 May 2018 amending Directive (EU) 2015/849 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing, and amending Directives 2009/138/EC and 2013/36/EU, OJ L 156.

the above-mentioned American model law drafted by the Uniform Law Commission.⁸⁹

The set of rules on consumer protection in crypto trading is dominated by information regulation. To a greater or lesser extent, there are also general provisions prohibiting dishonest or fraudulent acts or practices, in particular fraud or willful misrepresentation of material facts and market manipulation.

Among the obligations of pre-contractual information, the following dominate: the obligation to present data on service providers; the price of their use and the time frame of calculation; the user's right to be notified of changes in fees and other relevant elements relating to the transaction itself. In the case of investment-type cryptocurrencies, an abbreviated prospectus (the White Paper), as the equivalent to a more complex prospectus, is a condition for issuing financial instruments. Some regulations also provide mandatory warnings such as transfers irrevocability warnings, require the information on clients' rights in the event of a transfer error, and the right to proof of transaction execution. Information regulation in the phase of concluding the contract implies that the client has access to detailed data on the transaction and costs, but there are still no examples of detailed regulation of the contract itself.

More and more regulations stipulate an obligation, which is often a condition for granting a license, to establish a user protection policy which, among other, must include a procedure for resolving disputes, reporting unauthorized disposal of funds or errors; as well as informing the user about the manner of filing a complaint and the procedure for timely dispute resolution. However, there are opinions that the licensing requirements would not prove to be effective enough. In their recommendations for the cryptocurrency regulatory framework, the Central Bank of the Netherlands and the Financial Markets Authority have stated a number of reasons why they do not recommend the integration of obligations of service providers aimed at consumer protection, but suggest intensifying information campaign activities.

Among the arguments they stressed the fact that it would be difficult, almost impossible, to prevent cross-border trade in virtual space, given that in real space there is no international consensus on consumer protection standards that should be adopted in relation to cryptocurrency. On the other hand, the volatile value of virtual currencies and their speculative nature are an additional negative factor.⁹⁰

⁸⁹ Uniform Law Commission (n 77).

⁹⁰ De Nederlandsche Bank, AFM (n 23).

From a comparative perspective, the proposal for a Regulation on Markets in crypto-assets is an example of the most comprehensive specific standardization of rules on issuing, trading in crypto-assets and investor protection up to date. Particular emphasis is placed on the requirements regarding the publication of information, the prohibition of the use of insider information and other forms of market abuse. Requirements vary depending on the type of crypto-assets. The basic instrument of information regulation envisaged by the proposal of the Regulation on cryptocurrency is a Crypto-Asset White Paper (CAWP). Depending on the type of cryptocurrency in question, the Proposal set special requirements for publishing information about the offer, as well as the obligation of *ex ante* approval or notifications to the competent national supervisor. The most detailed information regulation obligations are provided for asset-referenced tokens. In addition to the general obligations of honest and professional conduct, the Proposal mandates a duty to communicate in a clear and unambiguous manner, advertising requirements (which must be in accordance with the mandatory data stated in the White Paper), as well as providing information on the value of issued coins and tokens, reserves and any event that may affect their value. The Proposal guarantees certain minimum rights to investors. Issuers of crypto-assets and related service providers are required to establish and maintain effective and transparent procedures for the prompt, fair and consistent resolution of investor complaints. European financial services regulatory agencies are expected to prescribe technical standards for dispute resolution procedures.

6. IN CONCLUSION: CHALLENGES FOR WESTERN BALKAN FINANCIAL REGULATORS

The above analysis, focused on the basic rationales for regulating the crypto market and basic principles on which the choice of regulatory strategy is based, should serve as a basis for the analysis of the challenges faced by the regulatory authorities in Western Balkan countries. Albeit the risks as a basis for regulating crypto-assets are very similar, the risk of violating national restrictions on capital movements and foreign exchange operations is more pronounced in these countries. Also, the bank-centered financial systems in this region and their leading role in the payment system is one of the additional factors that must be kept in mind. Over the past few years, fintech is a rising trend in South Eastern Europe.⁹¹ Still, not many regulators have decided to take big steps, while most of

⁹¹ See, for instance: Caroline Stern, *FinTechs and their emergence in banking services in CESEE* (2017) 3 Focus on European Economic Integration 17, 42.

them are aware of the developments and ready to prepare changes in money laundering regulations. Most regulators in Western Balkan countries are aware of the need to establish coordination between competent national authorities, create incentives in regulatory frameworks and adopt measures to encourage financial innovation. Overly concentrated banking sector, which predominantly owns payment systems, may provide a barrier to entry for non-bank financial services providers. Most WB countries have not yet implemented the pro-competitive requirements of the European Payment Services Directive aimed to improve access of smaller incumbents, which makes the regulatory response on fintech even more important. Given the shallow and underdeveloped financial market, where government bonds are the predominant financial instrument being traded, fintech is an opportunity for regulators to take measures to strengthen competition. To strengthen equity investments, it would be necessary to develop primary and secondary stock markets. For instance, in North Macedonia and Serbia Founderbeam, an Estonian fintech firm, enables trade and access to financial information on secondary markets through its platform.

In terms of principles underlying the design of a regulatory strategy, financial regulatory governance of NRAs in South-East Europe could be characterized as predominantly based on "command and control" and is characterized with a conservative focus based on rules, strong focus on controlling systemic risks as well as principal-agent relationship between regulators and regulated entities.⁹² Paradigm shift is one of the basic prerequisites for achieving a balanced approach to financial intermediation. Reforming the approach to regulation is a gradual process, regulators need to base their actions on proportionate regulatory tools to help mitigate regulatory failures. Therefore, phased access to designing regulatory strategy on crypto-assets could be based on successive steps.

As was the case in comparative regulatory practice, the first step of financial regulators in the Western Balkans was to disseminate information and issue warnings to the public, especially service users, and initiatives focused on raising awareness on the risks of cryptocurrencies and other crypto-assets. Market conduct supervisors of financial services and central banks warned on the volatility of crypto-currencies, operational risks such as the risk of cyberattack, risks stemming from money laundering activities, data protection etc.⁹³

⁹² Bajakić, 930.

⁹³The Bank of Albania:

<https://www.bankofalbania.org/Press/On_the_risks_associated_with_the_use_of_virtua

The following step is related to the interpretation of AML/CFT laws and changes to the legal framework to accommodate rules on new intermediaries, in particular crypto-asset exchanges (trading platforms) and providers of digital wallets, whereas mining activities are usually out of the regulatory focus. Here the regulators are confronted with the dilemma on whether applicable know-your-customer methods are allowed and under which conditions. Laws on AML/CFT in WB countries are mostly in line with the Fifth Anti-Money Laundering Directive. At minimum, providers of digital wallets and virtual currency exchange services should be subject to the AML/CTF rules. For instance, Montenegro transposed the Fifth Anti-Money Laundering Directive into Law on Prevention of Money Laundering and Terrorist Financing, establishing that natural and legal persons who are issuing or managing virtual currencies, as well as exchanging virtual currencies into fiat currencies, and vice versa, are subject to reporting duties. Serbian Law on Prevention of Money Laundering and Terrorist Financing extended the scope of legislation to include subjects providing services of purchasing, selling or transferring virtual currencies or exchanges of cryptocurrencies for money or other property through internet platform, devices in physical form or otherwise. The next step is to comply with the requirements of the amended FATF Recommendation No. 15 and ensure that crypto-asset service providers are subject to effective systems for monitoring and supervision, in line with a risk approach based on FATF guidance.⁹⁴

As all those countries aim to join the EU, in an area where there is no harmonized transnational regime, as well as binding regulation at EU level, regulators in the Western Balkans have the opportunity to demonstrate their creative discretion and residual legal powers. While the warnings mainly

l_currency.html>; The Republic of Srpska Securities Commission: <<http://www.secrs.gov.ba/Fajl.aspx?Id=579f9839-522c-4689-9eca-8a7522360b2f>>; The FbiH Securities Commission: <<http://www.secrs.gov.ba/Fajl.aspx?Id=579f9839-522c-4689-9eca-8a7522360b2f>>; The National Bank of Serbia: <<https://www.nbs.rs/internet/cirilica/scripts/showContent.html?id=7605&konverzija=no>>; <<https://www.nbs.rs/internet/cirilica/scripts/showContent.html?id=9604&konverzija=no>>; Serbian Securities Commission: <<http://www.sec.gov.rs/index.php/sr/index.php/sr/едукација/едукација-инвеститора/текстови/557-упозорење-улагачима-у-крипто-имовинска-права-криптовалуте-и-дигиталне-токене>>; The National Bank of North Macedonia: <http://nbrm.mk/ns-newsarticle-soopshtieniie_na_nbrm_28_9_2016.nsp> all accessed on 05 November 2020.

⁹⁴ FATF (n 2).

referred to virtual currencies, regulators in the observed countries did not comment on the issue of investment tokens. As noted in previous sections of this Paper, regulators in more developed countries are issuing guidance and developing financial instrument tests, to ensure that investment tokens are covered by existing capital market laws, in particular in relation to initial coin offerings. Up to date, none of the regulators in the WB issued official guidance on how existing securities laws are applicable to crypto-assets.⁹⁵

In the absence of more detailed guidelines from the regulator or rules adapted to new services, regulators in Western Balkan countries should depart from a sector based approach, in order to prevent legal uncertainty for firms experimenting with fintech, and promote competition based on innovation. The regulators could propose to extend the regulation of traditional financial services by stretching the interpretation of existing rules, to include risks arising from new services. As a previous or next step, in order to better understand the challenges faced by new financial intermediaries, to better understand the risks to the financial system, and thus propose better legislation regarding crypto-asset business, regulators and/or supervisors should embark on the journey of regulatory innovations. It refers to central banks, bodies for the supervision of financial services that can be consolidated under the roof of an agency or sector specific (e.g. securities commissions), ministries in charge of finance, as well as other institutions such as central registers and securities clearings, bodies in charge of preventing money laundering, protection of personal data etc. For instance, setting up an innovation office is a good way to create a first point of contact, especially when unclear regulatory framework deters incumbents to offer innovative products and services. Innovation offices, at least in theory, should be composed of skilled staff from different departments, able to identify regulatory issues.⁹⁶

⁹⁵ Securities Commission of the Republic of Serbia issued a short statement which states that crypto-asset activities are not free from regulatory burdens, and hence may fall under the scope of existing rules on capital markets, foreign exchange and anti-money laundering. A regulatory guidance is expected, for instance, in Republika Srpska, to follow the new Law on Capital Markets which will contain some articles on regulating investments in crypto-assets.

⁹⁶ For instance, the Ministry for Scientific and Technological Development, Higher Education and Information Society of the Republic of Srpska, is supporting the establishment of a Centre for Digital Transformation. One of the domains of interest of the Innovation Office in the North Macedonia, operational since June 2019, is the crypto-asset market, as well as the development of central bank digital currency. It aims to issue legally non-binding opinions relevant for the fintech market, scan the existing regulatory framework and develop a Fintech Strategy.

Creating a regulatory sandbox is another option to allow potential new market entrants or existing financial innovators to test financial products and services within a controlled environment, especially when the regulatory framework is non-existing or its application is unclear. To date, the first regulatory sandbox was established in Montenegro, where the Securities Commission adopted rules on its objective, application process and criteria for application. However, whether it really works and what are the effects is not yet known. As already mentioned in this Paper, not much should be expected from this regulatory option, as it implies a high level of competence of regulators and more significant resources.

Being diverse and involving a number of new intermediaries, crypto-asset activities, as a rule, are not regulated by single legislative act or specific regulatory regime. Bespoke regulatory regimes like Malta, France, Gibraltar are rather rare. Some of the Western Balkan financial regulators focused on most pressing issues, such is the prohibition of certain activities related to crypto trading and issues related to the risk of money laundering. For instance, in its warning, the National Bank of Serbia clarified that crypto-assets could not be used for making payment transactions, in line with provisions stipulating that national payment transactions should be executed, with a few exceptions, in dinar as a national currency. If issuing detailed guidance specifying under which conditions financial intermediaries are allowed to trade in crypto-assets, or clarifications on how to distinguish between intermediaries and activities in novel services resemble the traditional financial services did not yield results, regulators may consider regulating activities which fall outside the perimeter of existing regulations. This should be based on the level of activity in the given country, risks associated with services and the institutional capacities for enforcing new rules. It should be accompanied with clear rules on intertwined competences and responsibilities of multiple national regulators.

Creating a specific regulatory framework with rules tailored to address the risks of crypto-asset markets as a particular fintech sector is a two-edged sword. Financial innovations are rapidly evolving and prone to circumvent regulations. On the other side, stringent rules may overregulate the sector and raise the burden of compliance. Specific regulatory regime shifts the focus from sector based to activity based object of regulation and ensures that similar activities or functions are subject to the same rules, in line with the reasonable expectations of market participants.

To date, only Albania has adopted the bespoke legislation. The Law on Financial Market based on Distributed Ledger Technology was twice subject to the

parliamentary process, as the Prime Minister refused to sign it in June 2020, arguing that its application could cause financial distress. The law aims to cover different activities and intermediaries, especially activities of DLT exchanges and wallet providers, and include rules on ICO, with the focus on information regulation. The Law has defined investigation competences of the Albanian Financial Supervisory Authority. At the time of preparing this manuscript, Serbian Draft Law on Digital Property has been subject to public consultations, and is currently forwarded to the Serbian Government to decide on the official Draft which will enter the parliamentary procedure for adoption, together with amendments to the Law on AML/CFT. The latter aims to further harmonize Serbian AML legislation with the FATF standards, encompassing a wider range of tokens than cryptocurrencies. The main aim of this bespoke regime, which resembles to a certain extent the French crypto legislative framework, is to bring more clarity in relation to investment type of crypto-assets. Given the limitation of the remaining space in this Paper, as well as the fact that this is a draft law, only a few key points should be pointed out. The draft sets a clear intention of the legislator to legally differentiate digital property (and its forms) as a new institute, different from electronic money and digitalized securities, and designates the National Bank of Serbia and the Securities Commission as supervisory bodies (regulators) in the application of the future law. This means that every startup in Serbia will be able to raise capital by issuing digital tokens in accordance with the law, but at the same time, the legal regulation of the process will also protect the investors.

As the initial offer of digital assets is only the beginning of trade in digital tokens, the Draft also regulates the conditions for providers of services related to digital assets who, in order to obtain the permission of the supervisory body, must meet personnel, organizational and technical requirements. The draft also prescribes the conditions for secondary trading that must be performed through a platform organizer licensed to provide services related to digital assets as well as for OTC trading that the parties can perform without intermediaries.

To conclude, many developed countries are trying to regulate crypto-assets as the new form of financial assets by applying and expanding existing regulations in the field of capital markets and payment services.

On the other hand, some smaller countries strive, often by adopting special regulations and designing novel regulatory regimes, to create more liberal frameworks and position themselves in the international innovation market. After all, this can be seen in the Western Balkans, where Albania and Serbia are pioneers in the legislative initiative. Each regulatory approach has its advantages

and disadvantages. Time will show how effectively the new framework will be applied and whether its implementation will ensure the strengthening of competitiveness in the national financial market, as well as the strengthening of competitive advantages in relation to foreign markets.

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